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ODD END SEMESTER EXAMINATION DECEMBER- 2025

Name of the Course: B.Tech

Semester: III

Name of the paper: Automotive Systems

Paper Code: TME 312

Time: 3 hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory
- (ii) Answer any two sub questions among a, b, & c in each main question
- (iii) Total marks in each main question are twenty
- (iv) Each question carries 10 marks

Q1. 20 marks	
(a) Outline the major milestones in the global automotive industry.	CO1
(b) How has the automobile industry evolved in India since independence?	
(c) Write a short note on the importance of automobile applications in different industries.	
Q2. 20 marks	
(a) Explain the working and components of a multi-plate clutch system.	CO2
(b) What is Suspension? Explain any two types of suspensions with diagrams.	
(c) Discuss the function and necessity of the exhaust and intake systems in automobiles.	
Q3. 20 marks	
(a) Classify automobiles based on use and purpose with suitable examples.	CO3
(b) Explain the meaning of "Variants and Versions" in automotive design.	
(c) How do dimensions and space constraints influence vehicle design and architecture?	
Q4. 20 marks	
(a) Write a comparative note on internal combustion engines vs electric vehicles.	CO4
(b) Explain the term "Hydraulic Hybrid Vehicle". How does it work?	
(c) What are the limitations of current battery technology in EVs?	
Q5. 20 marks	
(a) Highlight the career opportunities available in the area of benchmarking in the automotive industry.	CO5, CO6
(b) Explain the importance of vehicle identification number (VIN) and nomenclature standards.	
(c) Discuss the economic and social impacts of automobiles in developing countries.	

